

- **LSTM**

Strengths: Effectively learns long-term temporal dependencies via gating.

Weaknesses: Large parameter footprint and sensitive hyperparameters, especially in sparse-data contexts.

- **Transformer**

Strengths: Captures long-range correlations efficiently with abundant data.

Weaknesses: Quadratic complexity in sequence length; memory-intensive.

- **Informer**

Strengths: Employs sparse self-attention to enhance scalability for long sequences.

Weaknesses: Multi-layer stacks involve thousands of parameters and require large datasets.

- **N-BEATS**

Strengths: Provides accurate forecasting with interpretable decompositions into trend and seasonal components.

Weaknesses: High parameter count and data-hungry; may overfit in scenarios with limited data.

- **MLP**

Strengths: Simple and straightforward to implement and train.

Weaknesses: Lacks specialized inductive biases for oscillatory or near-singular data.

- **RBF Network**

Strengths: Excels at local approximations and can model a broad class of functions.

Weaknesses: Placement of kernel centers is non-trivial; struggles with sharp rational spikes or steep gradients.

- **SIREN**

Strengths: Sinusoidal activations capture smooth and high-frequency signals effectively.

Weaknesses: Requires large capacity to model abrupt spikes or extremely limited data; not explicitly designed for missing-data imputation.

- **TCN**

Strengths: Uses dilated convolutions for efficient sequence modeling.

Weaknesses: Deeper architectures increase parameter counts and can be computationally slow.